

PROBLEM

Problem Statement – Question 1: Data Preparation & Inductive Learning

A healthcare company has patient records containing attributes such as age, blood pressure, cholesterol, glucose level, BMI, and family history. The objective is to prepare the data for diabetes prediction using an appropriate inductive learning approach. The target variable and important features must be identified. Since the dataset contains more non-diabetic than diabetic patients, the class imbalance must be handled using techniques such as SMOTE, under-sampling, or class weighting. The dataset should be divided into training and testing sets using a 70:30 ratio, followed by suitable preprocessing such as encoding and normalisation.

Problem Statement – Question 2: Decision Tree for Diabetes Diagnosis

Develop a Decision Tree classification model to predict whether a patient is diabetic or non-diabetic. The first three levels of the Decision Tree should be analysed, and the root node should be justified using Information Gain or Gini Index. The model should be evaluated using Accuracy, Precision, Recall, F1-Score, and False Negatives. Since false-negative predictions can result in missed diabetes cases, suitable methods for reducing false negatives should be discussed. Finally, post-pruning or cost-complexity pruning should be applied and compared with the original Decision Tree.

Problem Statement – Question 3: Statistical Learning for Risk Stratification

Develop a statistical learning model such as Logistic Regression or Naive Bayes for diabetes risk prediction. The statistical model should be compared with the Decision Tree using Accuracy and AUC-ROC. The system should identify the top three predictors of diabetes from both models and determine whether the important features identified by the two models agree. The results should also explain when a statistical learning model is preferable to a Decision Tree for healthcare risk prediction.

Problem Statement – Question 4: Reinforcement Learning for Treatment Recommendation

Design a Reinforcement Learning-based treatment recommendation system for diabetes patients. The problem should be formulated as a Markov Decision Process consisting of patient health states, possible treatment actions, rewards, and state transitions. The available actions may include Diet, Exercise, Medication, and Monitoring. A Q-Learning agent should be trained for at least five patient episodes and Q-table updates should be demonstrated for multiple state-action pairs across different iterations. The final learned policy should identify the best action for each patient state, while considering patient safety, bias, explainability, privacy, ethical deployment, and human clinical supervision.